CHAPTER 2

Introduction – Entity Analysis

Entity Analysis is the process of identifying the main entities, their attributes, primary keys, and relationships in a database. It helps in designing a well-structured database by organizing data efficiently, reducing redundancy, and ensuring data integrity. In the ArtisanHub e-commerce project, entities such as Customer, Artisan, Product, Orders, Payment, Delivery, Review, and Wishlist are analyzed to build an effective relational database.

Identify Entities, Attributes, Primary Keys, and Relationships

Entities and Attributes

Entity Name	Attributes	Primary Key
Customer	Customer_ID, Full_Name, Email, Phone_Number, Address, Password	Customer_ID
Artisan	Artisan_ID, Full_Name, Email, Phone_Number, Shop_Name, Address	Artisan_ID
Category	Category_ID, Category_Name, Description	Category_ID
Product	Product_ID, Product_Name, Price, Stock, Description, Category_ID, Artisan_ID	Product_ID
Inventory	Inventory_ID, Product_ID, Available_Stock, Last_Updated	Inventory_ID
Shopping_Cart	Cart_ID, Customer_ID, Product_ID, Quantity	Cart_ID
Orders	Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status	Order_ID
Order_Details	Order_Detail_ID, Order_ID, Product_ID, Quantity, Price	Order_Detail_ID

Payment	Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date	Payment_ID
Delivery	Delivery_ID, Order_ID, Delivery_Address, Delivery_Status, Expected_Date	Delivery_ID
Review	Review_ID, Customer_ID, Product_ID, Rating, Review_Text	Review_ID
Wishlist	Wishlist_ID, Customer_ID, Product_ID	Wishlist_ID
Coupon	Coupon_ID, Coupon_Code, Discount_Percentage, Expiry_Date	Coupon_ID
Coupon_Usage	Usage_ID, Coupon_ID, Customer_ID, Order_ID	Usage_ID

Primary Keys

Entity	**Primary Key**
Customer	Customer_ID
Artisan	Artisan_ID
Category	Category_ID
Product	Product_ID
Inventory	Inventory_ID
Shopping_Cart	Cart_ID
Orders	Order_ID
Order_Details	Order_Detail_ID
Payment	Payment_ID
Delivery	Delivery_ID
Review	Review_ID

Wishlist	Wishlist_ID
Coupon	Coupon_ID
Coupon_Usage	Usage_ID

Relationships

Entity 1	Relationship	Entity 2
Customer	Places	Orders
Customer	Adds	Shopping_Cart
Customer	Writes	Review
Customer	Creates	Wishlist
Artisan	Sells	Product
Category	Contains	Product
Product	Stored In	Inventory
Orders	Contains	Order_Details
Order_Details	Refers To	Product
Orders	Has	Payment
Orders	Has	Delivery
Product	Receives	Review
Coupon	Applied To	Orders

Customer	Uses	Coupon

Foreign Keys

Table	Foreign Key
Product	Category_ID, Artisan_ID
Inventory	Product_ID
Shopping_Cart	Customer_ID, Product_ID
Orders	Customer_ID
Order_Details	Order_ID, Product_ID
Payment	Order_ID
Delivery	Order_ID
Review	Customer_ID, Product_ID
Wishlist	Customer_ID, Product_ID
Coupon_Usage	Coupon_ID, Customer_ID, Order_ID

Conclusion

The identification of entities, attributes, primary keys, and relationships forms the foundation of the ArtisanHub database. A well-structured relational database design ensures that all information related to customers, artisans, products, orders, payments, deliveries, and reviews is organized efficiently. Defining primary and foreign keys helps establish accurate relationships between tables while maintaining data integrity and minimizing redundancy. This database design provides a strong framework for implementing SQL operations in the upcoming

phases of the project and supports the efficient management of the ArtisanHub e-commerce platform.